# PAPER_1312_TOP_YUKAWA

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# PAPER_1312 — Top Quark Yukawa y_t = 1 Naturalness

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**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
**Tier:** C — Particle Physics Open (CLOSED)
**Date:** June 11, 2026
**Calculator surface:** `calculate_paradox({"paradox": "top_yukawa"})`
**Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1312})`

## Master Expression

``

`y_t = m_t / (v/√2) = 1.0 natural (no fine-tuning)`
``

**Match: STRUCTURAL EXACT**

## Interpretation

Top sits at $v/\sqrt{2}$ minimum of Mexican-hat. $y_t = 1$ emerges naturally — no tuning required.

## UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{QCD} = 0.217$ GeV, $v_{Higgs} = 246$ GeV (PAPER_1270)

## NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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